
Styling & Layout

Introduction

- ▶ All components accept `style` prop, which can be:
 - ▶ A plain JavaScript object
 - ▶ A reference to a style created by `StyleSheet.create(...)`
 - ▶ Array of styles (last style has precedence)
- ▶ Applied to elements individually (no equivalence to CSS selectors)
- ▶ No `className` prop

Example

```
function App() {  
  return <View style={{ flex: 1 }}>  
    <Text style={styles.redText}>Text 1</Text>  
    <Text style={ [  
      styles.redText,  
      styles.grayBg,  
      { fontSize: 20 }  
    ]}>Text 2</Text>  
  </View>  
}
```

```
const styles = StyleSheet.create({  
  redText: {color: 'red' },  
  grayBg: { backgroundColor: 'gray' }  
});
```

Text

- ▶ Properties match how CSS works on the web
- ▶ Properties:
 - ▶ `color`
 - ▶ `fontFamily, fontSize, fontStyle, fontWeight, fontVariant`
 - ▶ `letterSpacing, lineHeight, textAlign, textAlignVertical`
 - ▶ `textShadowColor, textShadowOffset, textShadowRadius, textTransform`

Background

- ▶ Properties:
 - ▶ `backgroundColor`
- ▶ How to fill the background with an image?
 - ▶ `<ImageBackground>` component

Border

▶ Properties:

- ▶ `borderWidth, borderLeftWidth, borderTopWidth, ...`
- ▶ `borderColor, borderLeftColor, borderTopColor, ...`
- ▶ `borderRadius, borderTopLeftRadius, ...`
- ▶ `borderStyle`

Size

- ▶ Fixed dimensions

- ▶ `width`, `height` with numeric values (number of pixels)

- ▶ Flex dimensions

- ▶ `flex` with numeric values (ratio of taken space)

- ▶ Percentage dimensions

- ▶ `width`, `height` with percentage values (portion of parent)
 - ▶ `minWidth`, `maxWidth`, `minHeight`, `maxHeight`
 - ▶ Require parent with defined size

Flexbox Layout: Container Properties

- ▶ `flexDirection` (column, row, column-reverse, row-reverse): direction the children of a node are laid out
- ▶ `direction` (ltr, rtl): direction children and text in a hierarchy should be laid out
- ▶ `justifyContent` (flex-start, flex-end, center, space-between, space-around, space-evenly): how to align children within the main axis
- ▶ `alignItems` (stretch, flex-start, flex-end, center, baseline): how to align children along the cross axis
- ▶ `flexWrap` (wrap, nowrap, wrap-reverse): whether items are wrapped into multiple lines
- ▶ `alignContent` (flex-start, flex-end, center, stretch, space-between, space-around): distribution of lines along the cross axis when items are wrapped to multiple lines



Flexbox Layout: Item Properties

- ▶ `flex`
- ▶ `flexBasis`, `flexGrow`, `flexShrink`
- ▶ `alignSelf`: **like** `alignItems` **for container**
- ▶ `width`, `height`

- ▶ Flexbox playground:
 - ▶ <https://yogalayout.com/playground>

Position

- ▶ `position`: how an element is positioned within its parent
 - ▶ `relative`: element is offset to the normal position based on the values of `top`, `right`, `bottom`, `left`
 - ▶ `absolute`: element is positioned absolutely based on the values of `top`, `right`, `bottom`, `left`
- ▶ `zIndex`: components with larger value display on top of others

Overflow

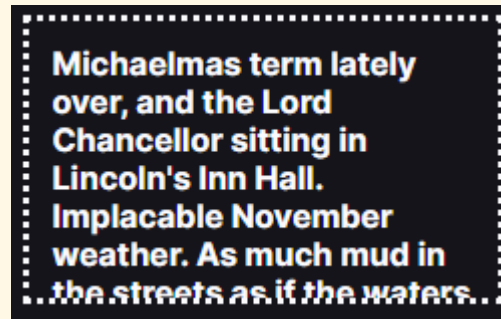
- ▶ `overflow`: how children are measured and displayed
 - ▶ `visible`: children not clipped, no scrollbar
 - ▶ `hidden`: children clipped
 - ▶ `scroll`: scrollbar shown

`visible`

A black rectangular box with a dashed white border containing white text. The text is longer than the box's height, causing it to overflow and be visible outside the bottom of the box.

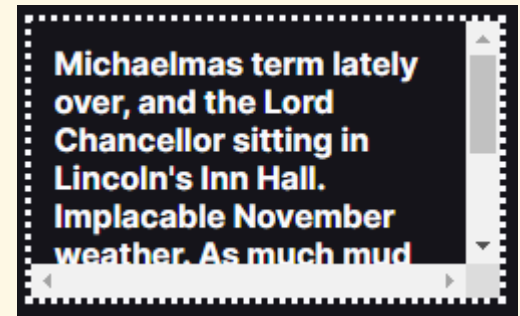
Michaelmas term lately
over, and the Lord
Chancellor sitting in
Lincoln's Inn Hall.
Implacable November
weather. As much mud in
the streets as if the waters...
had but newly retired from
the face of the earth.

`hidden`

A black rectangular box with a dashed white border containing white text. The text is longer than the box's height, but the bottom portion is clipped and not visible.

Michaelmas term lately
over, and the Lord
Chancellor sitting in
Lincoln's Inn Hall.
Implacable November
weather. As much mud in
the streets as if the waters...

`scroll`

A black rectangular box with a dashed white border containing white text. The text is longer than the box's height, and a vertical scrollbar is visible on the right side of the box.

Michaelmas term lately
over, and the Lord
Chancellor sitting in
Lincoln's Inn Hall.
Implacable November
weather. As much mud

StyleSheet class

- ▶ `create()` method
- ▶ `compose()` method
 - ▶

```
const App = () => <View style={lightApp}>
  ...</View>;
const page = StyleSheet.create({
  app: { ... },
});
const themes = StyleSheet.create({
  light: { ... },
});
const lightApp = StyleSheet.compose(
  page.app, themes.light);
```
- ▶ `flatten()` method

Transforms

- ▶ Allow to modify the appearance of components using 2D or 3D transformations
- ▶ Use `transform` style property
 - ▶ Supported transforms:
 - ▶ `rotateX`, `rotateY`, `rotateZ`, `rotate`
 - ▶ `skewX`, `skewY`
 - ▶ `translateX`, `translateY`
 - ▶ `scaleX`, `scaleY`, `scale`
 - ▶ `perspective`
 - ▶ `matrix`
- ▶ Example:
 - ▶

```
<Text style={{transform: [
  { scale: 2},
  { skewX: "45deg" },
  { rotate: "0.2rad" },
  { translate: -20 }
]}}>...</Text>
```

Exercises

- Implement the following screens:

